SUPPORTING ADULT GUIDANCE

Function notation, domain and range

Algebraic Expressions and Identities · Lesson 8. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This is the final lesson of the unit. In the main activity, learners choose one of four function machines ($2x - 1$, $x^2 + 3$, $1/(x - 2)$, $\sqrt{(x - 3)}$), feed it numbers and watch the arrows build a mapping diagram. Inputs that would divide by zero or take the square root of a negative number jam the machine and are listed as excluded. Each machine then asks questions: evaluate, solve, state the domain, describe the range. Learners then practise notation, read domain and range from graphs, and work as function detectives.

Driving Question: *What goes in, what comes out, and what is not allowed?*

The mathematics, in plain terms

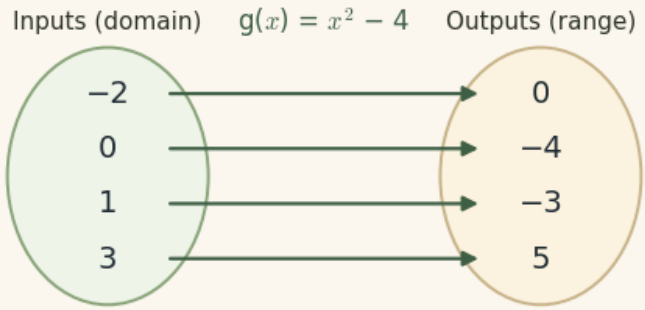
A function assigns to each element of its domain exactly one output. f: x ■ … and f(x) = … are equivalent notations. The domain is the set of allowed inputs (here: all real numbers, minus any value giving division by zero or the square root of a negative). The range is the set of outputs actually attained, e.g. $x^2 + 3$ has range f(x) ≥ 3 because $x^2 \geq 0$. The √ symbol denotes the non-negative root. Composite and inverse functions are not part of this lesson.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A mapping diagram and its range

g(x) = x^2 – 4 with domain {–2, 0, 1, 3}.

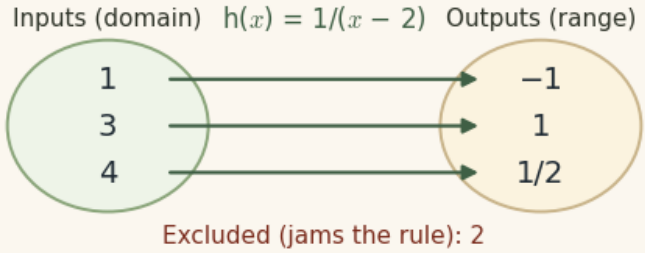


1. Feed in –2.	g(–2) = (–2)^2 – 4 = 0
2. Feed in 0 and 1.	g(0) = –4, g(1) = –3
3. Feed in 3.	g(3) = 9 – 4 = 5
4. Collect the outputs: the range.	{–4, –3, 0, 5}

Range: {–4, –3, 0, 5}. Each input has exactly one arrow, so g is a function.

Example 2 — An input that jams the machine

h(x) = 1/(x – 2). Which input must be excluded? Can h(x) equal 0?



1. Feed in 1, 3 and 4.	h(1) = –1, h(3) = 1, h(4) = 1/2
2. Feed in 2: you cannot divide by zero.	1/(2 – 2) = 1/0
3. So exclude 2 from the domain.	domain: all numbers except 2
4. If 1/(x – 2) = 0, multiply by (x – 2).	1 = 0, which is false
5. So 0 never comes out.	0 is not in the range

Domain: all numbers except 2. The output 0 can never appear.

Machine, then picture, then graph

Learners experience the machine first: feeding numbers and seeing jams makes 'domain' concrete before it is named. The mapping diagram shows 'one output per input' visually, and the graph explorer then connects the same ideas to the axes: domain left to right, range up and down. The proof stretch asks why some outputs can never appear.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The function machine and mapping diagram: evaluate, solve, domain, range. Stretch: why some outputs never appear.	~16 min
Movement	Notation moves with f(x) = 3x + 4.	~6 min
Reflection	Domain and range shaded on four graphs.	~8 min
Creativity	Function detective: find the rule; is it a function?; design your own.	~10 min
Rest	A machine that takes almost anything — except one number.	~2 min

Nutrition	Intellectual: knowing the limits of a rule.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Reading f(3) as f × 3.
Ask: “What does the machine do when you feed in 3?”

Writing f(*a*) as a number, or leaving x in the answer.
Ask: “What replaces x this time?”

Thinking two inputs with the same output stop it being a function.
Ask: “Does any single input have two arrows leaving it?”

Excluding the value that makes the numerator zero.
Ask: “Which part of a fraction can never be zero?”

Giving the range of $x^2 + 3$ as all numbers.
Ask: “Can a square ever be negative? So what is the smallest output?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substitution, solving a linear equation, and knowing that you cannot divide by zero or take the square root of a negative number. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says the domain of $1/(x - 2)$ is “$x > 2$”. The aim is a question that lets them test the claim themselves.

- “Feed in 1. Does the machine jam?”

- “Which input actually jams it?”
- “What happens just either side of 2?”
- “So which numbers are allowed?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are composite and inverse functions covered?
No. This lesson builds the notation, domain and range they will rely on later.

Why is the domain of $\sqrt{(x-3)}$ $x \geq 3$ and not $x > 3$?
Because $\sqrt{0} = 0$ is allowed. Learners can test it by feeding 3 into the machine.

Is the stretch required?
No. The secure core is evaluating, solving, and finding the domains and ranges in the machines.

What comes next?
This completes the Algebraic Expressions unit. A unit check-in or the next unit follows.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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